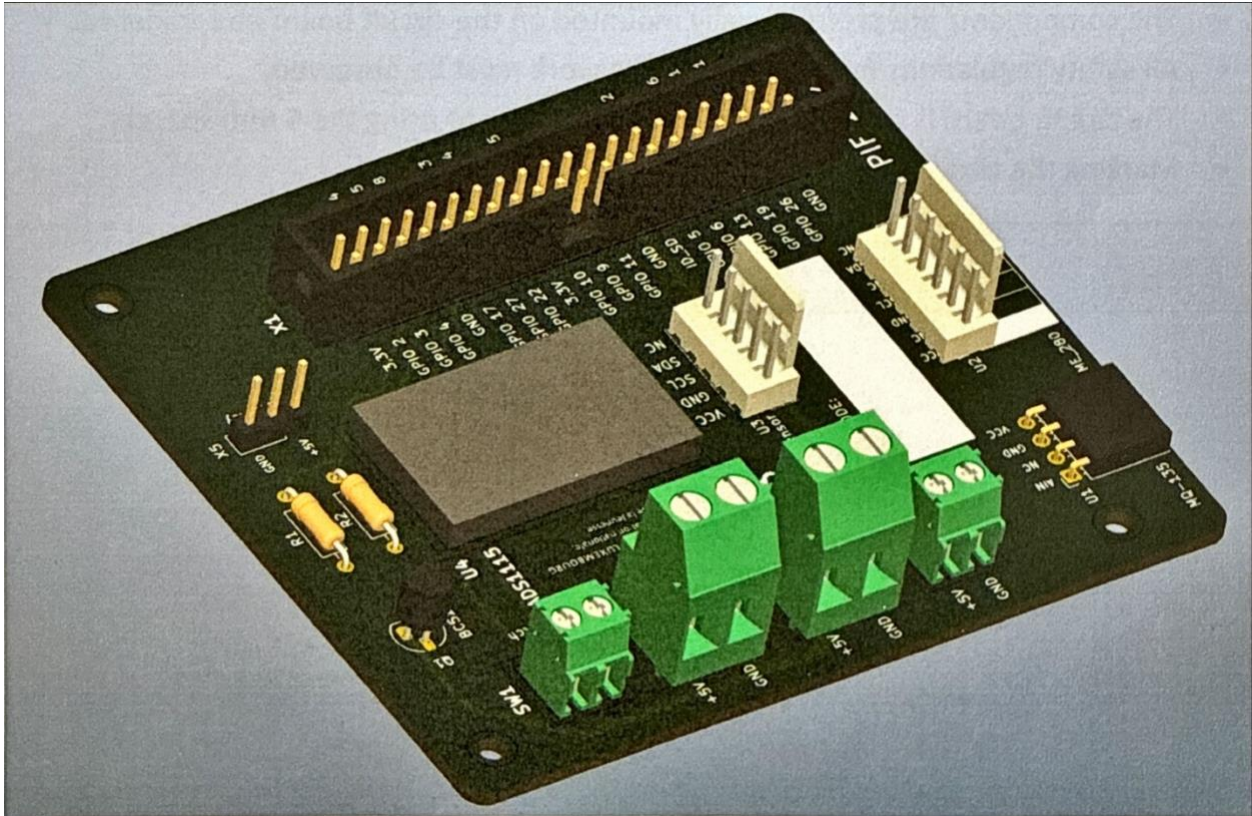


Userstory A2

SOFRA Jonathan



Printed Circuit Board

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Requirements according to the work order

In this userstory, we only have 3 requirement that needs to be done. (“Must” Requirement)

- The necessary components are soldered onto the circuit board.
- The circuit board is screwed onto the PVC base plate
- Documentation:
 - o Work steps
 - o List of materials

List of materials

Connectors

- **40-pin Raspberry Pi GPIO header** (female, right-angle)
- **X3 – Power_OUT_PI**
 - 2-pin screw terminal (GND / +5V)
- **X4 – Power_IN**
 - 2-pin screw terminal (GND / +5V)
- **X1 – Switch**
 - 2-pin screw terminal
- **X5 – Display**
 - 3-pin
- **X2 – LED Strip**
 - 2-pin DC plug connector (barrel-jack style)
- **U1 – MQ-135 gas sensor connector**
 - 4-pin female header (AIN / NC / GND / VCC)
- **U2 – BME280 connector**
 - 4-pin JST-XH: VCC, GND, SCL, SDA
- **U3 – GY-302 (BH1750) light sensor connector**
 - 4-pin JST-XH: VCC, GND, SCL, SDA
- **Display connector**
 - 3-pin header (GND / +5V / Signal)

Integrated Circuits & Semiconductors

ADS1115 (16-bit ADC module, through-hole header)

Q1 – BC517 NPN transistor

Work steps

1. Preparation

Gather all materials from the Bill of Materials (BOM).

Verify that the PCB matches the revision shown in the documentation.

Set up your workspace with:

- Soldering iron + stand
- Solder (preferably lead-free)
- Tweezers
- Side cutters
- Multimeter for continuity testing
- Inspect the PCB for:
 - Manufacturing defects
 - Clean solder pads
 - Proper silkscreen labeling for each component

2. Install Low-Profile Components First

(Always solder from lowest to highest components.)

2.1 Resistors

- Insert the resistors in the designated places (R1, R2, etc.).
- Bend the leads on the backside to hold them in place.
- Solder the pads.
- Cut off the excess leads.

3. Install Semiconductor Components

3.1. Q1 – BC517 Transistor

- Orient the transistor according to the silkscreen (flat side alignment).
- Insert the leads into the PCB.
- Solder the pads and trim the leads.

3.2. ADS1115 ADC Module

- Solder a row of **male pins** onto the ADS1115 board.
- Insert the module into the appropriate footprint (U4).
- Solder each pin from the underside of the main PCB.

4. Install Connectors

4.1. Raspberry Pi 40-Pin Header

- Place the **right-angle female 40-pin header** into the GPIO footprint.
- Ensure the connector is perfectly aligned and flush with the PCB.
- Solder all 40 pins on the back.

4.2. Screw Terminals

Install in the following order:

- **X4 – Power_IN (2-pin)**
- **X3 – Power_OUT_PI (2-pin)**
- **X1 – Switch (2-pin)**
- **X5 – 3-pin**

Steps:

- Insert each terminal block.
- Ensure orientation matches the silkscreen.
- Solder all pins.

4.3. Sensor & Peripheral Connectors

U1 – MQ-135 Connector (4-pin female header)

Insert the 4-pin female header.

Keep it perpendicular to the board.

Solder all pins.

U2 – BME280 Connector (4-pin JST-XH)

Follow the same steps as above.

U3 – GY-302 Light Sensor Connector (4-pin JST-XH)

Same procedure.

X5 – Display Connector (3-pin header)

Insert → align → solder.

4.4. LED Strip Connector (X2 – 2-pin DC Barrel Jack)

Insert the DC jack into position.

Confirm it sits flat on the PCB.

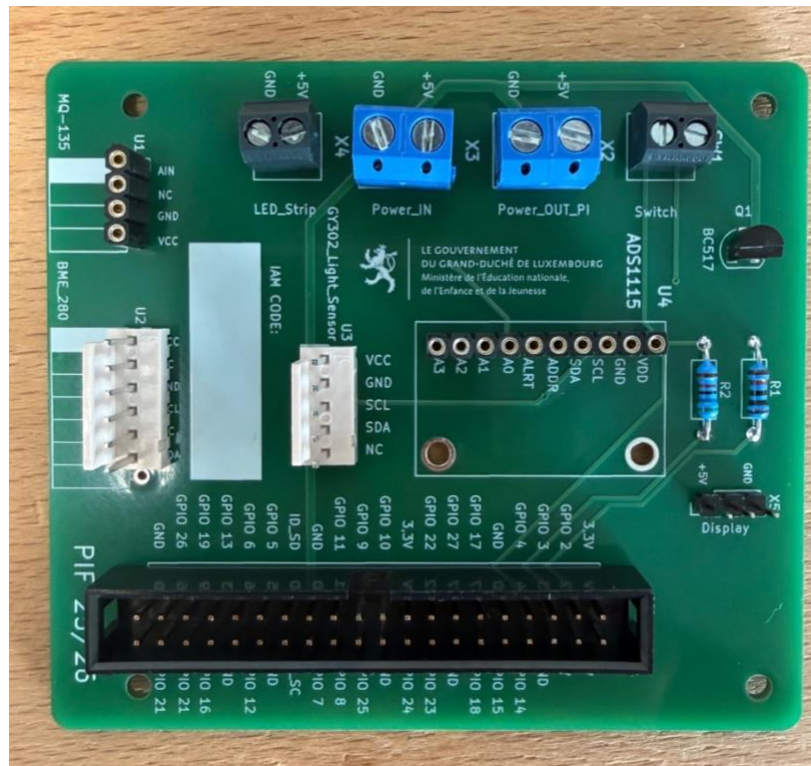
Solder the large mechanical pads first, then the signal pins.

5. Final Assembly

Mount the PCB to its case or enclosure using the four mounting holes.

Insert all cable harnesses and ensure correct polarity.

Document firmware version or Raspberry Pi OS configuration if relevant.



Here you can see the final step of this user story, having the touch display, the raspberry pi and our PCB mounted and screwed onto our PVC holder.

